How Do You Learn? — An Interactive Learning & Quantum Computing Survey

We are a student team developing an interactive learning platform for Quantum Computing and Quantum Algorithms.

This survey aims to understand how students, educators, researchers, and professionals currently learn technical concepts, the challenges they face, and what features could make learning Quantum Computing more engaging and effective.

Your responses will help us design a platform that better addresses real learning needs. The survey is anonymous and takes approximately 5 minutes to complete.

* Indicates required question

SECTION 1 -- About You

1. Q1. Which describes you right now? *

Mark only one oval.

- ◯ High school student (Class 9–12)
- ◯ Undergraduate student (B.Tech / B.E. / B.Sc. / BCA / etc.)
- ◯ Postgraduate student (M.Tech / M.Sc. / MBA / etc.)
- ◯ PhD scholar / researcher
- ◯ Faculty / teacher (school, college, coaching, online)
- ◯ Working professional
- ◯ Other: _____

2. Q2. Your institution type *

Students: your college/school. Professionals: your employer.

Mark only one oval.

- National-level institute
- State / government university or college
- Private university / engineering college
- School (K–12)
- Research institute / lab
- Company / industry
- Other: _____

3. Q3. If you're a working professional or faculty — years of experience *

Mark only one oval.

- Under 2
- 2–5
- 5–10
- 10+
- N/A — I'm a student
- Other: _____

4. Q4. If you're a working professional — your sector *

Mark only one oval.

- IT / software / SaaS
- Education / edtech
- Research & development
- Finance / consulting
- Core engineering (electronics, manufacturing, energy, etc.)
- N/A — I'm a student / faculty
- Other: _____

Section 2 -- How you learn

Not about quantum yet — just how you learn anything technical. **Be honest**, not polite.

5. Q5. Which of these have you actually used to learn something? *

Check all that apply.

- [ ] YouTube
- [ ] Coursera / edX / Udemy
- [ ] NPTEL / SWAYAM
- [ ] Duolingo
- [ ] Brilliant
- [ ] Khan Academy
- [ ] Physics Wallah / Unacademy / other Indian edtech apps
- [ ] Codecademy / freeCodeCamp / HackerRank / LeetCode
- [ ] Kaggle / Google Colab / Jupyter notebooks
- [ ] ChatGPT / Gemini / other AI chatbots
- [ ] None of these
- [ ] Other: _____

6. Q6. Be honest — what makes you open a learning app again on day 15?

Check all that apply.

- [ ] It feels like a game (streaks, XP, levels)
- [ ] Instant feedback — I know immediately if I'm right or wrong
- [ ] Lessons are bite-sized (5–10 minutes)
- [ ] I can see my progress visibly
- [ ] It adapts to my level
- [ ] Good explanations when I'm stuck
- [ ] Friends / community / competition
- [ ] Certificates or credentials
- [ ] I never make it to day 15
- [ ] Other: _____

7. Q7. What ruins an online course for you? *

Check all that apply.

- [ ] Hours of passive video, no doing
- [ ] Quizzes that don't test real skill
- [ ] No help when I get stuck
- [ ] Certificate at the end, no actual skill gained
- [ ] Can't practice hands-on
- [ ] Content outdated or shallow
- [ ] Too expensive
- [ ] I lose motivation halfway
- [ ] Constant upselling / ads
- [ ] Other: _____

8. Q8. How do you prefer learning a genuinely hard technical topic?

Mark only one oval.

- ◯ Build and try things myself, with instant feedback
- ◯ Watch videos first, then try
- ◯ Read structured notes / diagrams
- ◯ Ask an AI chatbot as I go
- ◯ Classroom with a teacher
- ◯ Solve graded challenges to level up
- ◯ Other: _____

9. Q9. Your comfort with programming *

Mark only one oval.

- Comfortable — I can write code from scratch
- Can read and modify code, can't write from scratch
- Beginner — school level / a little
- None — I don't code

10. Q10. If you've used AI chatbots to learn something technical — how trustworthy were they? *

Mark only one oval.

- Mostly accurate
- Mostly accurate, but I double-check everything
- They confidently made mistakes — I've caught them
- I don't trust them for learning
- Never used AI to learn

Section 3 -- Quick Check: Quantum

Quantum computing = next-gen computers that use physics instead of plain 1s and 0s.

Whatever your level — total beginner to expert — there are questions for you. That's the point of the question below.

11. Q11. Which one describes you with quantum computing? *

Mark only one oval.

- Quantum what? / Just heard the word → Novice *Skip to question 12*
- Know roughly what it is, never tried learning → Novice *Skip to question 12*
- Tried learning, gave up midway → Learner *Skip to question 14*
- Currently learning it → Learner *Skip to question 14*
- Already know it well — use it in study / work / research → Expert *Skip to question 16*

NOVICE SECTION

You're exactly who we're building for. Honesty helps more than politeness here.

12. Q12. What's kept you from exploring quantum computing so far? *

Mark only one oval.

- Don't know where to start
- Sounds too difficult — "for geniuses only"
- Assumes heavy math
- No time / other priorities
- Don't see the career relevance yet
- No fun or interactive way to learn it — just lectures and PDFs
- My institution doesn't offer anything
- Honestly, just not curious

13. Q13. If a platform existed where you drag-and-drop quantum gates in your browser and see results instantly — no math needed, nothing to install, works on any laptop — would you try it?

Mark only one oval.

- Yes, sounds fun
- Maybe — if it's genuinely beginner-friendly
- Only if my college / course included it
- Probably not — not my area

Skip to question 19

LEARNER SECTION

You've already explored quantum computing, so we want to understand what you tried, where you struggled, and what could make the learning experience better.

14. Q14. What have you used so far to learn quantum computing? *

Mark only one oval.

- YouTube videos
- MOOCs (Coursera / edX / Udemy)
- NPTEL / SWAYAM
- IBM Quantum / Qiskit tutorials
- Textbooks
- College course / lectures
- Interactive tools (Quirk, IBM Quantum Composer)
- AI chatbots
- Reddit / Discord / communities

15. Q15. What's been hardest? *

Mark only one oval.

- Concepts too abstract (superposition, entanglement, interference)
- Math prerequisites
- Too theoretical — not enough hands-on
- Simulator setup pain (installing, environments)
- The tools themselves have steep learning curves
- No feedback when I'm stuck
- No structured, step-by-step path
- Motivation — I dropped once already

Skip to question 19

EXPERT SECTION

You're rare. We need you.

16. Q16. Tools you've actually used *

Mark only one oval.

- Qiskit
- Cirq
- PennyLane
- Braket / qBraid
- Quirk / IBM Quantum Composer
- Microsoft Quantum Katas
- Nothing formal — papers and textbooks only

17. Q18. What's missing in existing quantum learning tools? *

Mark only one oval.

- No gentle on-ramp for absolute beginners
- Assumes math maturity from page one
- No instant-feedback, build-and-see circuit interaction
- No AI assistance / debugging help
- Poor visualization of quantum states
- No auto-graded assessments
- Frameworks are siloed — Qiskit, Cirq, PennyLane live in separate worlds
- Nothing — existing tools are good enough

18. Q19. Would you recommend an interactive platform like this to students you teach or mentor? *

Mark only one oval.

- Yes, definitely
- Maybe — depends on quality
- No
- N/A — I don't teach or mentor anyone

Skip to question 19

FINAL

Thank You for staying till the end. Your feedback is valuable to us !!!!

19. Q20. A browser-based platform to learn quantum computing the Duolingo way — interactive, instant feedback, AI tutor. How interested are you?

Mark only one oval.

	1	2	3	4	5	
Zero	◯	◯	◯	◯	◯	Sign me Up

20. Q21. If your institution gave you access to this platform, how likely would you be to use it?

Mark only one oval.

- ◯ Definitely
- ◯ Probably
- ◯ Only if it counted toward my grades
- ◯ Unlikely

21. Q22. Faculty only: if this platform had auto-graded quantum labs and a dashboard showing exactly where each student is stuck, would you use it in a course or elective?

Mark only one oval.

- ◯ Yes, definitely
- ◯ Maybe — if it maps to my syllabus
- ◯ No
- ◯ N/A — I'm not faculty

22. Q23. Last one — in a line or two: what would actually make you try (or keep) learning quantum computing? Please add your feedback *

23. We'd love to hear your thoughts. If you have any suggestions, feedback, or ideas that could help us improve the platform, please share them below.

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